

SMART IRRIGATION USING SOIL-MOISTURE SENSING AND LoRa

Review 2 — System Design (revised)

Final Year Project 2025–26

Submitted by

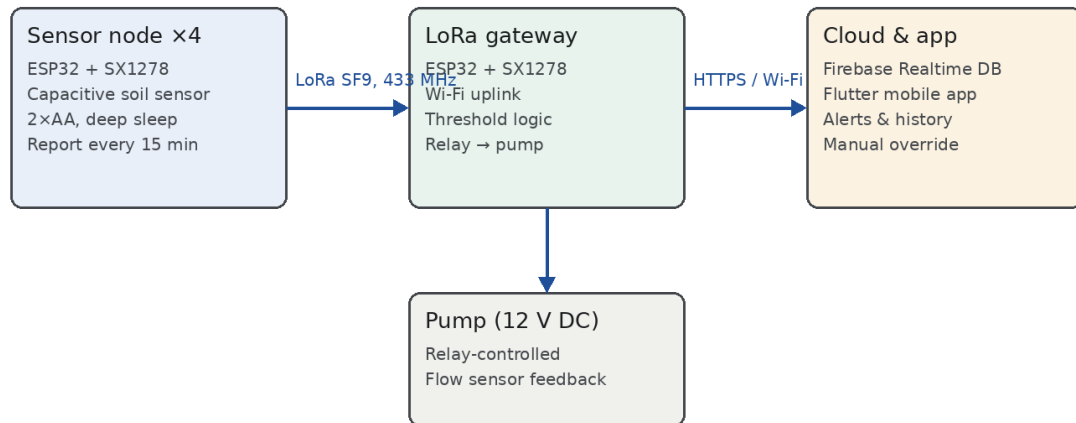
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1. System architecture

Fig. 2.1 System architecture — Smart Irrigation (LoRa)



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Figure 2.1 shows the three tiers of the system: battery-powered sensor nodes, a mains-powered gateway, and the cloud dashboard.

2. Module description

Module	Hardware / software	Responsibility
Sensor node	ESP32, SX1278, capacitive probe	Sample moisture, transmit, sleep
Gateway	ESP32, SX1278, relay	Receive, decide, switch pump
App	Flutter, Firebase	Live readings, history, alerts

3. Power budget (added after review)

State	Current	Time per cycle
Deep sleep	10 μ A	899 s
Sensor read	40 mA	0.5 s
LoRa TX (SF9, 17 dBm)	120 mA	0.4 s

Average current ≈ 0.09 mA. With 2xAA cells (2500 mAh), the expected life is about 14 months.